Sadeeq Adeyemi

3D perception and multi-object tracking under occlusion

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RESEARCH INTERESTS

Multi-object tracking and segmentation (MOTS) through occlusion; 3D perception from cameras and lidar; state estimation and data association; video foundation models (SAM 3); evaluation on public benchmarks.

EDUCATION

MS, Data Science and Engineering, University of Maine, Orono, USA — 2025 – present

MS, Structural Engineering, University of Maine, Orono, USA — 2023 – 2025

MSc, Project Management, University of Portsmouth, United Kingdom — 2023

BEng, Civil Engineering, Federal University of Technology Akure, Nigeria — 2019

RESEARCH EXPERIENCE

Graduate Researcher, CVAR Lab (Computer Vision and Robotics), University of Maine — 2026 – present

How much identity can a tracker keep while an object is hidden? I designed and ran every experiment below.

- Benchmarked SAM 3 and SAM 3.1 with official HOTA: KITTI-MOTS val cars 82.6 and 74.6; MOTS20 train 72.6 and 61.5.
- Defined an occlusion measure from MOT17 visibility labels: SAM 3 kept identity through 94% of 245 hides, SAM 3.1 63%.
- Traced about half of SAM 3.1's deficit to its shipped tracking settings by reading the source code; a controlled run raised its car HOTA from 74.0 to 78.3, against SAM 3's 82.3. Isolated a separate fault that halts it on long, crowded sequences.
- Built a Kalman-filter identity layer on SAM's output: it repaired SAM 3.1 (identity through occlusion 63% to 70%, McNemar $p = 0.004$) but not SAM 3, which showed that motion outside the tracker says where an object is, not who it is.
- Mined 10,871 pedestrian occlusions in 1,256 NVIDIA driving clips with a 3D line-of-sight test. On that geometry an occlusion-aware tracker kept 57% of identities through 1 to 5 second hides; a standard tracker kept 2%.
- Built a simulation lab for controlled occlusion experiments (46 automated tests): identity kept rose from 0% to 89% as visibility reasoning, negative information and re-identification were added; MathWorks JPDA reached 57%.

Robotics Research Assistant, CVAR Lab, University of Maine — 2025 – present

- Built mobile robots end to end, from sensor integration and wiring to ROS 2 nodes and lidar point-cloud perception.
- Implemented EKF localization fusing IMU, wheel odometry and lidar, with costmap planning and obstacle avoidance.

RESEARCH OUTPUTS

- Technical report: *Solving occlusion in MOTS: SAM 3 and SAM 3.1 zero-shot*, CVAR Lab, October 2026. Available on request.
- Code: SAM 3 / SAM 3.1 benchmark, evaluation and filter layer, github.com/sadeeqgandalf/av-stack.
- Code: occlusion-aware 3D tracker in C++ (track existence from measured detection rates and sight lines; KITTI val association accuracy 77.7 vs AB3DMOT's 76.6 for cars, 0.15 ms per frame), github.com/sadeeqgandalf/occlusion-tracker.
- Code and live lab: occlusion-aware tracking, github.com/sadeeqgandalf/lunar-occlusion-tracking.

INDUSTRY EXPERIENCE

Materials Engineer, R.W. Gillespie, Maine — 2024

Machine Learning Engineer and Structural Steel Fabricator, NewTech Steel, Nigeria — 2018 – 2023

- Built and deployed a computer vision model for real-time fatigue and defect detection in steel, halving inspection time.

Civil Engineering Intern, Federal Capital Development Authority, Abuja, Nigeria — Feb – Aug 2017

- Worked on the construction of wastewater treatment facilities for Nigeria's Federal Capital Territory.

TECHNICAL SKILLS

Methods: multi-object tracking, Kalman filtering (EKF, UKF), data association (Hungarian, JPDA, MHT), 3D geometry and ray casting, re-identification

Software and data: Python, C++17, PyTorch, ROS 2, OpenCV, filterpy, MATLAB, TrackEval; KITTI, KITTI-MOTS, MOTS20, MOT17, NVIDIA PhysicalAI-AV

Certifications: AWS Certified Machine Learning Specialty; AWS Certified Solutions Architect, Associate